

20250410高代习题课

Recall that V, U are linear space on number field K . $\varphi: V \rightarrow U$ is a linear isomorphic map, if

- (1) φ is a bijection
- (2) $\varphi(\alpha + \beta) = \varphi(\alpha) + \varphi(\beta)$, for any $\alpha, \beta \in V$.
- (3) $\varphi(k\alpha) = k\varphi(\alpha)$, for any $\alpha \in V, k \in K$.

1. A, B are $m \times n$ -dimensional matrix on number field K . U and V are the solution spaces of homogeneous linear equations $Ax=0$ and $Bx=0$, respectively.

Prove that if $\text{rank } A = \text{rank } B$, then $\exists$ invertible T_{nm} on K , such that

$f(y) = Ty$ ($\forall y \in U$) is a linear isomorphic map from U to V .

2. V, V' are finite-dimensional linear spaces of number field K

$\phi: V \rightarrow V'$, a linear map, i.e., $\phi(\alpha + \beta) = \phi(\alpha) + \phi(\beta)$ for any $\alpha, \beta \in V, k \in K$.
 $\phi(k\alpha) = k\phi(\alpha)$

Prove that $\exists$ direct sum decomposition:

$$V = U \oplus W, \quad V' = M \oplus N, \quad \text{s.t. } \text{Ker } \phi = U, \quad W \cong M \text{ (isomorphic)}$$

3. C is the complex field. Let $V = \left\{ A = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix} \mid \alpha, \beta \in C \right\}$.

Clearly, V is a linear space on real number field R regarding matrix addition, and multiplication of numbers and matrices.

(1) find a basis of V , and $\dim V$

(2) If $\dim V = n$, find an isomorphic map $\varphi: V \rightarrow R^n$.

4. It is known that $\alpha_1 = (1, -1, 4)$, $\alpha_2 = (1, 0, 2)$, $\alpha_3 = (0, 3, -5)$ form a basis of R^3

Let V^* be a dual space of R^3 . Find the dual basis of V^* for $(\alpha_1, \alpha_2, \alpha_3)$.